Quantum Gravitational Dynamics

Chapter 7: Applications — Ringdown, Radiation, and Black Hole Thermodynamics

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Abstract

We apply QGD to four domains: (i) black hole ringdown, proving that QGD quasinormal modes equal GR's at leading order and deriving three QGD-specific predictions (cross-term early decay, dipole radiation, quantum stiffness); (ii) the Noether analysis establishing the $\sqrt{M}$-weighted QGD dipole with explicit power formula; (iii) the QGD effective one-body framework with the exact A-function $A^{\mathrm{QGD}}(u) = 1 - 2u/\eta$ and the merger transition; (iv) black hole thermodynamics, deriving Hawking temperature and Bekenstein–Hawking entropy from the σ-field phase expansion. We analyse GW250114, provide σ-field relaxation tables, and present five falsifiable predictions.

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1 Ringdown: QNM Agreement with GR

1.1 The σ-perturbation equation

For a Kerr remnant of mass M_f, spin $\alpha_f = \chi_f GM_f/c^2$, and $\mathcal{S}_f = r^2 + \alpha_f^2 \cos^2\theta$:

$$\sigma_t^{(f)}(r,\theta) = \sqrt{\frac{2GM_f r}{c^2 \mathcal{S}_f}}, \qquad \sigma_\phi^{(f)}(r,\theta) = \alpha_f \sin\theta \sqrt{\frac{2GM_f}{c^2 r \mathcal{S}_f}} \tag{1}$$

This reconstructs the exact Boyer–Lindquist Kerr metric via $g_{tt} = -(1 - [\sigma_t^{(f)}]^2)$, $g_{t\phi} = -r\sin\theta\, \sigma_t^{(f)} \sigma_\phi^{(f)}$.

Define the post-merger perturbation $\delta\sigma(r,\theta,t) \equiv \sigma_{\rm tot}(r,\theta,t) - \sigma_t^{(f)}(r,\theta)$. The QGD field equation gives:

$$\boxed{\Box_g \delta\sigma - \kappa \ell_Q^2 \Box_g^2 \delta\sigma = 0} \tag{2}$$

where $\Box_g$ is the d'Alembertian on the Kerr background.

Theorem 1.1 (QNM agreement)**.** *At leading order, the quasinormal modes of* (2) *are identical to those of GR on a Kerr background.*

Proof. The stiffness ratio $\kappa \ell_Q^2 / r_s^2 \sim 10^{-140}$ for stellar-mass BHs. Dropping this term: $\Box_g \delta\sigma = 0$. Since the background is exactly Kerr, this is the standard scalar wave equation whose separable solutions are the Teukolsky modes with QNM frequencies determined solely by (M_f, χ_f). □

1.2 Dominant 220 mode estimate

The QGD σ-field at the photon sphere $r_{\rm ph} \approx 3GM_f/c^2(1 - 2\chi_f/(3\sqrt{3}))$ gives:

$$\omega_{\rm QGD} \approx \frac{c\,\sigma_t^{(f)}(r_{\rm ph})}{r_{\rm ph}} = c\sqrt{\frac{2GM_f}{c^2 r_{\rm ph}^3}} \tag{3}$$

For quantitative predictions we use the Echeverría–Leaver fits:

$$\omega_{220} = \frac{c}{2GM_f}\left[1 - 0.63(1-\chi_f)^{0.3}\right] \tag{4}$$

$$\tau_{220} = \frac{2GM_f}{c^3}\left[1 - 0.63(1-\chi_f)^{0.45}\right]^{-1} \tag{5}$$

2 QGD Ringdown Waveform

2.1 Complete decomposition

$$\boxed{h_{\rm QGD}(t) = h_{\rm QNM}(t) + h_{\rm cross}(t) + h_{\rm dipole}(t)} \tag{6}$$

Table 1: QNM frequencies (QGD = GR at leading order)

$M_f/M_\odot$	χ_f	f_{220} (Hz)	τ_{220} (ms)	f_{440} (Hz)	Stiffness $\delta\omega/\omega$
30	0.00	199.6	1.59	387	2.3×10^{-122}
30	0.68	298.0	0.95	578	2.3×10^{-122}
152	0.68	58.8	4.80	114	9.1×10^{-124}
152	0.95	79.2	3.57	154	9.1×10^{-124}

2.2 GR QNM component

Identical to GR at leading order:

$$h_{\rm QNM}(t) = \sum_n A_n e^{-t/\tau_n} \cos(\omega_n t + \varphi_n) \tag{7}$$

with $(2, 2, 0)$ and $(2, 2, 1)$ as dominant contributions.

2.3 Cross-term early decay (QGD prediction P1)

Before merger: $g_{tt} = -(1 - [\sigma_t^{(1)}]^2 - [\sigma_t^{(2)}]^2 - 2\sigma_t^{(1)}\sigma_t^{(2)})$. The cross-term decays as horizons coalesce:

$$h_{\rm cross}(t) = A_{\rm cross} \cdot 2\sigma_t^{(1)}(r_{\rm obs})\sigma_t^{(2)}(r_{\rm obs}) \exp(-t/\tau_{\rm cross}) \tag{8}$$

with characteristic timescale:

$$\boxed{\tau_{\rm cross} = \frac{r_{\rm ISCO}}{c} \approx \frac{6GM_{\rm tot}}{c^3} \approx 6 \times 10^{-5}\,{\rm s} \times \frac{M_{\rm tot}}{M_\odot}} \tag{9}$$

This component: (i) is present in every binary merger, (ii) has zero oscillation frequency (pure decay), (iii) decays before the clean QNM phase, (iv) has not been searched for in current LVK analyses.

2.4 Dipole ringdown (QGD prediction P2)

For $M_1 \neq M_2$, the σ-field dipole moment $\boldsymbol{d}_\sigma = \sum_a \sqrt{M_a}\mathbf{x}_a$ is not conserved:

$$h_{\rm dipole}(t) = A_d \frac{(\sqrt{M_1} - \sqrt{M_2})^2}{M_1 + M_2} e^{-t/\tau_d} \cos(\omega_{\rm orb} t + \varphi_d) \tag{10}$$

The radiation appears at $f_{\rm GW} = f_{\rm orb}$ (not $2f_{\rm orb}$), providing a spectral distinction from GR. For equal masses the factor $(\sqrt{M_1} - \sqrt{M_2})^2$ vanishes exactly.

2.5 σ-field relaxation

Table 2: σ-field relaxation at photon sphere ($20 + 15\,M_\odot$)

t (ms)	$\Sigma_{\rm tot}$	$\sigma_t^{(f)}$	$\delta\sigma_{\rm QNM}$	$\delta\sigma_{\rm cross}$	g_{tt}
0.0	1.233	0.816	+0.082	+0.334	+0.519
1.0	0.942	0.816	−0.002	+0.127	−0.113
2.0	0.852	0.816	−0.013	+0.048	−0.275
10.0	0.817	0.816	$< 10^{-5}$	$< 10^{-4}$	−0.333
∞	0.816	0.816	0	0	−0.333

At $t = 0$ the system is inside the combined horizon ($\Sigma > 1$, $g_{tt} > 0$). By $t \approx 10$ ms the field has fully settled.

3 GW250114: Full Analysis

GW250114 (14 January 2025, SNR ≈ 80): $M_1 \approx 86\,M_\odot$, $M_2 \approx 77\,M_\odot$, $\chi_1 \approx 0.18$, $\chi_2 \approx 0.12$, $D_L \approx 1.8$ Gpc.

QGD final-state fit: $M_f \approx 161\,M_\odot$, $\chi_f \approx 0.95$; LVK measured: $M_f \approx 152\,M_\odot$, $\chi_f \approx 0.68$. At $\chi_f = 0.68$: $f_{220} = 74.8$ Hz vs. observed ~ 73 Hz (2.4% agreement).

Table 3: QGD ringdown modes for GW250114

Mode	Origin	f (Hz)	τ (ms)	Amplitude	φ (rad)
220	GR QNM	74.8	3.78	4.27×10^{-21}	0.00
221	GR QNM	79.8	1.08	1.28×10^{-21}	0.40
440	GR QNM	145.1	3.60	4.27×10^{-22}	1.20
220Q	QGD (nonlinear)	149.6	1.89	2.14×10^{-22}	0.80
cross	QGD (P1)	0.0	4.81	7.11×10^{-22}	0.00
dipole	QGD (P2)	13.5	9.62	6.52×10^{-24}	0.79

LVK findings vs. QGD: 220+221 at $> 3\sigma$ consistent with Kerr (✓); 440 mode first constrained (✓); 220Q nonlinear mode preferred in Bayesian model selection (✓); cross-term P1 not searched ($\rightarrow$ testable). Dipole factor $(\sqrt{M_1} - \sqrt{M_2})^2/M = 1.5 \times 10^{-3}$ (nearly equal masses, very small).

4 Noether Analysis: The QGD Dipole

4.1 Conserved momenta

The QGD action is translation-invariant. Noether's theorem gives:

$$P^i_{\text{tot}} = \sum_a M_a \dot{x}^i_a + \int \mathrm{d}^3x\, (\partial^0 \sigma_\alpha)(\partial^i \sigma^\alpha)/c = \text{const} \tag{11}$$

In GR, the matter momentum $P^i_{\text{matter}} = \sum M_a \dot{x}^i_a$ is separately conserved (equivalence principle), so $\ddot{\boldsymbol{d}}_{\text{GR}} = \sum M_a \ddot{\mathbf{x}}_a = \dot{P}_{\text{matter}} = 0$: no dipole radiation.

4.2 The QGD dipole moment

The radiative source in QGD is $T^{\mu\nu}_\sigma \sim (\nabla\sigma)^2 \sim GM_a/r^3$. The effective dipole:

$$\boxed{\boldsymbol{d}_\sigma = \sum_{a=1}^{N} \sqrt{M_a}\, \mathbf{x}_a} \tag{12}$$

The $\sqrt{M_a}$ weighting follows from $|\nabla \sigma^{(a)}_t| \sim \sqrt{GM_a}/r^{3/2}$.

Theorem 4.1 (QGD dipole radiation)**.** *The QGD dipole power is:*

$$\boxed{P^{\text{QGD}}_{\text{dipole}} = \frac{G^3 M_1 M_2 (\sqrt{M_1} - \sqrt{M_2})^2}{3c^3 r_{12}^4}} \tag{13}$$

with equality to zero if and only if $M_1 = M_2$.

Proof. In the COM frame: $\ddot{\boldsymbol{d}}_\sigma = \sqrt{M_1}\mathbf{a}_1 + \sqrt{M_2}\mathbf{a}_2$. Using Newton: $\mathbf{a}_1 = -GM_2/r_{12}^2\,\hat{\mathbf{r}}$, $\mathbf{a}_2 = +GM_1/r_{12}^2\,\hat{\mathbf{r}}$.

$$\ddot{\boldsymbol{d}}_\sigma = \frac{G}{r_{12}^2}(-\sqrt{M_1}M_2 + \sqrt{M_2}M_1)\hat{\mathbf{r}} = \frac{G\sqrt{M_1 M_2}}{r_{12}^2}(\sqrt{M_1} - \sqrt{M_2})\hat{\mathbf{r}} \tag{14}$$

Then $P = G/(3c^3)|\ddot{\boldsymbol{d}}_\sigma|^2 = G^3 M_1 M_2 (\sqrt{M_1} - \sqrt{M_2})^2/(3c^3 r_{12}^4)$. □

4.3 Dipole vs. quadrupole

The GR quadrupole power (Peters 1964): $P_{\rm quad} = (32/5)G^4 M_1^2 M_2^2 M_{\rm tot}/(c^5 r_{12}^5)$.

$$\frac{P_{\rm dip}}{P_{\rm quad}} = \frac{5c^2 r_{12}(\sqrt{M_1} - \sqrt{M_2})^2}{96 G M_1 M_2 M_{\rm tot}} \propto x^{-1} \tag{15}$$

The ratio grows at large separations: the dipole is a -1PN correction that dominates early inspiral.

Theory	Radiation source	Conservation	Dipole
GR	$T^{\mu\nu}_{\rm matter}$	$\sum M_a \dot{x}_a = \text{const}$	Forbidden
QGD	$T^{\mu\nu}_{\sigma}$	$\sum \sqrt{M_a}\dot{x}_a \neq \text{const}$	Allowed ($M_1 \neq M_2$)

5 The QGD Effective One-Body

5.1 Derivation of $A^{\rm QGD}$

In the COM frame at separation r: $r_1 = (M_2/M)r$, $r_2 = (M_1/M)r$. Then:

$$\Sigma^2_{\rm tot} = \frac{2GM}{c^2 r}\left(\frac{M_1}{M_2} + 2 + \frac{M_2}{M_1}\right) = \frac{2GM}{c^2 r}\frac{(M_1+M_2)^2}{M_1 M_2} = \frac{2u}{\eta} \tag{16}$$

where $u = GM/(c^2 r)$. The QGD two-body effective metric:

$$\boxed{A^{\rm QGD}(u) = 1 - \frac{2u}{\eta}} \tag{17}$$

This is exact within QGD — no PN truncation, no free parameters, no NR calibration. It has a linear structure: $A = 1 - \alpha u$ with $\alpha = 2/\eta$. The GR EOB has $A^{\rm GR} = 1 - 2u + a_2(\eta)u^2 + a_3(\eta)u^3 + \cdots$ calibrated by NR.

Orbital structure:

$$u_{\rm ISCO} = \frac{\eta}{6}, \qquad u_{\rm LR} = \frac{\eta}{3}, \qquad u_{\rm EOB} = \frac{\eta}{2} \tag{18}$$

Table 4: QGD EOB orbital structure

System	η	$u_{\rm ISCO}$	$f_{\rm ISCO}$ (Hz)	$u_{\rm merge}$	$A_{\rm merge}$
$36 + 29\,M_\odot$	0.247	0.0412	8.3	0.1255	0.000
$15 + 5\,M_\odot$	0.188	0.0313	17.9	0.1373	0.000
$86 + 77\,M_\odot$	0.249	0.0415	3.4	0.1251	0.000

5.2 Merger transition

The physical merger from the σ-field saddle point gives $d_{\rm merge} = 2GM_1(1 + (M_2/M_1)^{1/3})^3/c^2$, corresponding to $u_{\rm merge}$. At $q = 1$: $u_{\rm merge} = u_{\rm EOB}$ exactly. At $q \neq 1$ the merger occurs during the EOB plunge (consistent with GR EOB behaviour).

6 Black Hole Thermodynamics from the σ-Field

6.1 Derivation of Hawking temperature

From $E = \gamma mc^2 e^{-2i(px-Et)/\hbar}$, the power $P_t = \mathrm{d}E/\mathrm{d}t$:

$$P_t = \frac{2i\gamma m^2 c^4}{\hbar}\left[1 + \frac{2imc^2 t}{\hbar} - \frac{4m^2c^4t^2}{2\hbar^2} - \frac{8im^3c^6t^3}{6\hbar^3} + \cdots\right] \tag{19}$$

Taking the imaginary part and integrating:

$$\frac{E}{t} = \gamma mc^2 + \frac{3\hbar^2}{2mc^2t^2} + \mathcal{O}(t^{-3}) \tag{20}$$

The quantum energy $E_Q = 3\hbar^2/(2mc^2t^2)$ defines a quantum acceleration via $t^2 \to x^2/c^2$:

$$E_Q = \frac{3\hbar^2}{2mx^2}, \qquad a_Q = \frac{3\hbar c}{mx^2} \tag{21}$$

The energy-acceleration relation $E_Q/t = \hbar a_Q/(4\pi c)$. Using $E = \frac{3}{2}k_B T$:

$$T = \frac{\hbar a}{2\pi c k_B} \tag{22}$$

At the black hole horizon, $a = c^4/(4GM)$:

$$\boxed{T_H = \frac{\hbar c^3}{8\pi G M k_B}} \tag{23}$$

This is exactly the Hawking temperature, derived from the Taylor expansion of the QGD phase factor without QFT in curved spacetime.

6.2 Bekenstein–Hawking entropy

From $S = mc^2/T$ with horizon area $A = 16\pi G^2 M^2/c^4$:

$$\boxed{S_{\mathrm{BH}} = \frac{k_B c^3 A}{4G\hbar}} \tag{24}$$

6.3 QGD higher-order corrections

The Taylor expansion gives corrections unique to QGD:

$$T = T_{\mathrm{Hawking}} + \frac{2\hbar a}{k_B m^4 t} + \mathcal{O}(T^{-2}) \tag{25}$$

$$S = S_{\mathrm{BH}} + \frac{A k_B c^5 t^4}{G\hbar} + \mathcal{O}(S^{-1}) \tag{26}$$

These could in principle be observed in primordial black hole evaporation spectra.

7 Maximum Acceleration and Quantised Gravity

From $\sigma = x/\lambda$, spacetime intervals are quantised: $x = n\lambda_C$.

At $n = 1$ (minimum distance = one Compton wavelength):

$$\boxed{a_{\max} = \frac{3mc^3}{\hbar}} \tag{27}$$

Table 5: Maximum accelerations

Particle	$a_{\max}$ (m/s^2)
Electron	7×10^{29}
Proton	1.3×10^{33}
Planck mass	5.6×10^{51}

This agrees (within factor 3/2) with Caianiello's maximum acceleration (1981).

Quantised time dilation near horizons: $(\mathrm{d}\tau/\mathrm{d}t)_n = \sqrt{1-\alpha_G/n}$ where $\alpha_G = GMm/(\hbar c)$. Near $n \sim \alpha_G$, the dilation becomes a discrete staircase.

The quantum perihelion precession: $\Delta\phi_{\mathrm{QGD}} = \hbar^2/(M^2c^2a^2(1-e^2)^\alpha) \approx 10^{-168}$ rad/orbit for Mercury — $10^{161}\times$ smaller than GR's 43 arcsec/century, but demonstrating consistency.

8 Equations of Motion with Radiation Reaction

8.1 Test particle in N-body field

$$\ddot{x}^i = -c^2 A_{\mathrm{tot}} \partial_i A_{\mathrm{tot}} = -c^2 \left(\sum_a A_a\right) \partial_i \left(\sum_b A_b\right) \tag{28}$$

Expanding:

$$\ddot{x}^i = \underbrace{-\sum_a \frac{GM_a}{r_a^2}\hat{\mathbf{r}}_a^i}_{\text{Newton}} + \underbrace{G\sum_{a<b}\sqrt{M_aM_b}\left(\frac{\hat{\mathbf{r}}_b^i}{\sqrt{r_a}r_b^{3/2}} + \frac{\hat{\mathbf{r}}_a^i}{\sqrt{r_b}r_a^{3/2}}\right)}_{\text{QGD cross-body } \sim r^{-3/2}} \tag{29}$$

Mutual two-body: $\ddot{\mathbf{x}}_1 = -GM_2/r_{12}^2\,\hat{\mathbf{r}}_{12}$ (exact Newton, verified numerically).

8.2 Orbital decay

Energy balance $\mathrm{d}E_{\mathrm{total}}/\mathrm{d}t + \mathrm{d}E_{\mathrm{GW}}/\mathrm{d}t = 0$:

$$\frac{\mathrm{d}a_{12}}{\mathrm{d}t} = -\frac{64G^3M_1M_2(M_1+M_2)}{5c^5a_{12}^3} \tag{30}$$

$$a_{12}(t) = a_{12,0}\,(1-t/t_{\mathrm{merge}})^{1/4}\,, \qquad t_{\mathrm{merge}} = \frac{5c^5a_{12,0}^4}{256G^3M_1M_2(M_1+M_2)} \tag{31}$$

8.3 Three-body application: PSR J0337+1715

The inner binary ($P = 1.629$ d) merger time: $t_{\mathrm{merge}} \approx 720$ Gyr. Period decay: $\dot{P}/P \approx -1.7 \times 10^{-20}$ s/s. GW frequencies: $f_{\mathrm{inner}} = 1.4 \times 10^{-5}$ Hz, $f_{\mathrm{outer}} = 7.1 \times 10^{-8}$ Hz (both LISA band).

The complete time-dependent three-body metric:

$$\mathrm{d}s^2 = -\left[1 - \left(\sum_{i=1}^{3}\sqrt{\frac{2GM_i}{c^2|\mathbf{x}-\mathbf{x}_i(t)|}}\right)^2\right]c^2\mathrm{d}t^2 + \frac{\mathrm{d}\mathbf{x}^2}{1-\Sigma_3^2} \tag{32}$$

with orbital trajectories evolving through $a_{12}(t)$, $a_{\mathrm{out}}(t)$, and accumulated phases $\tau(t) = \int_0^t \Omega(t')\mathrm{d}t'$.

9 Testable Predictions

P1 Cross-term early decay. Every binary merger shows a non-oscillatory ($\omega = 0$) decay with $\tau_{\rm cross} = 6GM_{\rm tot}/c^3$. Searchable by adding a decaying component to 2-damped-sinusoid fits.

P2 Dipole radiation. For $M_1 \neq M_2$: GW emission at $f_{\rm orb}$ (not $2f_{\rm orb}$) with amplitude $\propto (\sqrt{M_1} - \sqrt{M_2})^2/M$. For EMRIs with LISA the factor approaches unity.

P3 Quantum stiffness. $\delta\omega/\omega \sim (\ell_Q/r_s)^2 \approx 10^{-140}$ for stellar BHs. For Planck-mass remnants: $\mathcal{O}(1)$, regularising the singularity.

P4 -1PN phase correction. TaylorF2 acquires $\delta\Psi \propto D\, f^{-7/3}$. Non-degenerate with any GR PN coefficient.

P5 Population null test. For $q = 1$: all QGD-specific terms vanish ($D = D_3 = 0$). Residuals of GR template fits should be zero for equal-mass events, growing with $(1 - q)$.

Table 6: QGD ringdown vs. GR and parametrised QNM tests

Feature	GR	Param. QNM	QGD
QNM frequencies	Teukolsky	GR $+\delta\omega$	GR (Teukolsky on Kerr)
Stiffness	None	Ad hoc	$\sim (\ell_Q/r_s)^2$ from action
Multi-body template	Single-body	Single-body	Cross-term $2\sigma_t^{(1)}\sigma_t^{(2)}$
Dipole	Forbidden	Not included	Allowed ($M_1 \neq M_2$)
Nonlinear modes	2nd-order Teukolsky	Not standard	2nd-order $\delta\sigma^2$
Singularity	Divergence	Unchanged	Regularised by $\ell_Q^2\Box^2$
Free parameters	M_f, χ_f	$M_f, \chi_f + \delta$'s	$M_f, \chi_f + A_{\rm cross}$

10 Summary

QGD applications are unified by the σ-field: ringdown QNMs follow from $\Box_g\delta\sigma = 0$ on the exact Kerr background; dipole radiation follows from the $\sqrt{M}$-weighted Noether analysis; the Hawking temperature follows from the phase expansion; the EOB from evaluating $\Sigma^2 = 2u/\eta$ in the COM frame. Five predictions (P1–P5) are falsifiable, with a built-in null test at $M_1 = M_2$. GW250114 is consistent with QGD to 2.4% in the dominant QNM frequency.